

M1 — Rotation + polygon mass properties

Two gates, **both done** — all bit-for-bit vs the real original Nape AS3 (`m1-rotation.test.ts` , 3 tests). M1a = rotation + central impulse (circles); M1b = polygon mass/inertia.

Tested vs the **real original Nape AS3** (harness `tools/nape-oracle/harness-m1.as` → FFDec → Ruffle → golden), same as M0. See the [version rule](#).

M1a — rotation + central impulse

Extends the contact-free engine with angular dynamics and `applyImpulse` , verified on circles (no `sin/cos` in the traced values, so still exact).

Replica additions (`nape-core.ts`): `setVel` , `setAngVel` , `getInertia` , and `applyImpulse` (central: `vel += J·imass` , `Body.as:2406`). The angular integrator (`updateVel sinertia` branch) and rotation path (`updatePos` , `dr = angvel·dt`) were already ported in M0; M1a is the first time they're exercised.

Gate (`m1-rotation.test.ts` , bit-for-bit vs original):

- **SPIN** — a circle given `angularVel = 5.0` under gravity: mass + inertia, then 180 steps of (`y` , `vy` , `rot` , `angvel`) . Confirms angular drag decay (`5.0 → 4.780`), rotation accumulation, circle inertia (`32.572 = 72·area·density`), and that linear free-fall is unaffected by spin.
- **KICK** — a circle given a central `applyImpulse(37, -53)` : `vx = 81.79 = 37/mass` , then 60 steps of (`x` , `y` , `vx` , `vy`) .

✓ SPIN: mass, inertia and 180 steps (`y` , `vy` , `rot` , `angvel`) match the original
✓ KICK: central `applyImpulse` + 60 steps (`x` , `y` , `vx` , `vy`) match the original

sin/cos note: `updatePos` calls `Math.sin/cos` for the body axis once a body rotates, but the *traced* values (`rot` is pure $\sum \text{angvel} \cdot dt$) don't depend on it for a centred body — so M1a is bit-exact across AVM2/V8. The trig only reaches observable state once a rotation feeds geometry (offset COM, or rotated vertices in collision) — first relevant in M3.

M1b — polygon mass/inertia

Replica additions (`nape-core.ts`): `addPolygon` + a `validateShapeGeom` helper porting the polygon `area / inertia` (`ZPP_Polygon.as:1198`) and `localCOM` centroid (`ZPP_Body.as:360`) — the shoelace traversals, in Nape's exact vertex order (`cur = v1...v_{n-1}` , `v0`); `align` extended to shift polygon vertices.

Robustness: the harness traces the polygon's *actual stored vertices* (post Nape construction) and the test feeds those exact verts to the replica — so any winding/normalisation Nape applies can't cause a mismatch. (For the box, Nape kept the input order.)

Gate (`POLY` in `m1-rotation.test.ts`): a 40x24 convex box, `angularVel = 3.0` , under gravity:

- `mass = 0.96` (= `area·density/1000`) and `inertia = 174.08` (= `m/12·(w2+h2)`) match the original bit-for-bit;
- 120 steps of (`y` , `vy` , `rot` , `angvel`) match.

✓ POLY: box mass, inertia and 120 steps (y, vy, rot, angvel) match the original

Convex decomposition stays out of the bit-exact loop — the harness feeds an already-convex polygon, as the replica's `addPolygon` expects.

Deferred: off-centre `applyImpulse` (angular term) and offset-COM `align` shift — not needed until later milestones; will be verified when first exercised.